# Validation did not confirm development-only selections

Report 2 of 3 · Validation and frozen selection

Per-coin P&L and joint basket alpha/betas, with confidence intervals, for all nine frozen strategies. Cash-excess performance at 3% nominal.

## Executive Summary

- **All four development-selected candidates lost in 2024–2025**, with net returns from **–22.31% to –45.31%** at 20 bps. Their positive development Sharpes did not generalize to the next two years.
- The broad development-Q1 group retained a useful relative ordering, but not positive median economics: its validation median return was **–16.27%** and **30.73%** of cases were profitable. “Q1” is the best approximately 20% of development Sharpe ranks within each basket, not a calendar quarter.
- A separate validation-informed rule produced **five Track B candidates** from **39,332 eligible pairs**. Their validation Sharpes of **1.14–1.85 are selection statistics**, not independent evidence of discovery. B02 had no eligible candidate.
- The final research shortlist preserves the **original four plus five Track B selections**. No single combined portfolio, weights or winner is implied.

### Read the periods correctly

**Development: 1 January 2022–31 December 2023.** Design and diagnose the rule using this period only.
**Validation: 1 January 2024–31 December 2025.** Test Track A unchanged, then explicitly label later validation-informed selection as Track B.

The reserved evaluation was subsequently approved for **1 January–30 September 2026**, with an October 1 UTC closing endpoint. Its outcomes belong only in Report 3. Performance and factor diagnostics use the audited saved strategy paths and existing factor prices; signals, trades and selections are fixed.

**Primary convention:** 20 bps one-way fees on full-L1 target changes. Lag1 means idealized observed-close execution, not a real one-hour delay.

# 1. Track A tested development robustness, not the validation winner

The original development-only screen required positive 2022 and 2023 returns and Sharpes, positive full-development returns under 40 bps and an additional-hour execution delay, and more than half of both parameter and all 28 one-swap basket neighbors to be profitable.

The robustness score was the minimum of the two annual Sharpes and the two neighborhood median Sharpes. At most one case-basket was selected per family; score ties used lower turnover, experiment ID and basket ID. No family gate was relaxed when empty.

The rule was designed after development robustness had been inspected. Freezing it prevents later reselection; it does not remove development search bias or earlier historical inspection.

### The same four identities in both periods

| ID | Period | Net return | Sharpe | Annual L1 turnover | Max drawdown |
|---|---|---|---|---|---|
| A01 | Development | +165.36% | 1.84 | 31.44× | 23.25% |
| A01 | Validation | -29.55% | -0.39 | 35.15× | 44.82% |
| A02 | Development | +252.94% | 2.03 | 49.58× | 58.37% |
| A02 | Validation | -39.63% | -0.64 | 55.43× | 47.29% |
| A03 | Development | +64.38% | 1.19 | 27.62× | 19.79% |
| A03 | Validation | -22.31% | -0.39 | 22.34× | 43.11% |
| A04 | Development | +314.22% | 1.43 | 50.50× | 51.35% |
| A04 | Validation | -45.31% | -0.22 | 63.91× | 70.96% |

All four failed the economic expectation of positive net validation returns. The result is not simply that they failed to remain rank one. Original risk also persisted: A04's validation maximum drawdown reached **70.96%**. The four stay in the record unchanged rather than being replaced by favorable controls.

## 2. The reference controls isolate a basket comparison

“Reference” means the **predefined liquidity-tier reference basket**, ID 0: BTC, ETH, SHIB, BNB, AVAX, DOGE, FIL and GALA. It is a fixed comparison, not a claim that this basket is optimal.

Each row compares exactly the same strategy settings on the selected basket and reference basket. The selected baskets had much stronger development results. In validation, reference momentum and long/short trend were slightly profitable, while their selected-basket versions lost. Reversal was worse on the reference basket, so there is no universal reference advantage.

### Same settings, selected versus reference basket

| ID | Period | Selected net | Reference net | Selected Sharpe | Reference Sharpe |
|---|---|---|---|---|---|
| A01 | Development | +165.36% | +8.19% | 1.84 | 0.28 |
| A01 | Validation | -29.55% | -9.90% | -0.39 | -0.06 |
| A02 | Development | +252.94% | -13.26% | 2.03 | -0.10 |
| A02 | Validation | -39.63% | +8.66% | -0.64 | 0.29 |
| A03 | Development | +64.38% | +12.63% | 1.19 | 0.35 |
| A03 | Validation | -22.31% | -38.60% | -0.39 | -0.73 |
| A04 | Development | +314.22% | +98.46% | 1.43 | 0.86 |
| A04 | Validation | -45.31% | +5.45% | -0.22 | 0.35 |

A basket substitution changes every cross-sectional rank and hence other coins' positions; trend models retain fixed per-slot budgets. This is a matched strategy comparison, not a causal coin-removal experiment. Validation-profitability of a control is evidence to retain and investigate, not permission to silently promote it.

## 3. The basket comparison also changes risk

The same-pair comparison must retain turnover and drawdown, not just returns and Sharpe. These rows use the identical strategy settings, dates, 20 bps fee convention and idealized observed-close execution as the preceding table.

Annual TO is full-L1 target-change turnover in times per year. Maximum drawdown is the positive loss magnitude from the saved continuous two-year path. This table is a comparison of realized paths, not a risk-adjusted portfolio or a causal basket treatment.

Reference risk was not uniformly lower. A03 reversal had a larger validation drawdown on the reference basket (**53.46%**) than on the selected basket (**43.11%**), despite slightly lower turnover. A04's reference drawdown was lower, but still **58.37%**.

### Same settings, selected versus reference risk

| ID | Period | Selected annual TO | Reference annual TO | Selected max DD | Reference max DD |
|---|---|---|---|---|---|
| A01 | Development | 31.44× | 32.69× | 23.25% | 35.33% |
| A01 | Validation | 35.15× | 33.52× | 44.82% | 30.83% |
| A02 | Development | 49.58× | 52.75× | 58.37% | 44.13% |
| A02 | Validation | 55.43× | 50.74× | 47.29% | 22.08% |
| A03 | Development | 27.62× | 24.00× | 19.79% | 32.31% |
| A03 | Validation | 22.34× | 21.41× | 43.11% | 53.46% |
| A04 | Development | 50.50× | 54.25× | 51.35% | 56.16% |
| A04 | Validation | 63.91× | 65.29× | 70.96% | 58.37% |

# 4. Strong development paths depended on period-specific positions

The four candidates were profitable in both development years by construction. Validation exposed a different pattern: A01 recovered in 2025 after a deep 2024 loss; A02–A04 lost in both validation years. Two-year windows and annual views answer different questions, so they are shown together rather than moving window boundaries after seeing results.

### Annual net returns on continuous paths

| ID | 2022 | 2023 | 2024 | 2025 |
|---|---|---|---|---|
| A01 | +66.26% | +59.60% | -39.15% | +15.77% |
| A02 | +140.95% | +46.47% | -32.43% | -10.66% |
| A03 | +32.05% | +24.48% | -1.91% | -20.80% |
| A04 | +83.54% | +125.69% | -13.10% | -37.06% |

### Legacy LUNA slot: saved net-P&L attribution

| ID | Development LUNA contribution (pp) | Validation LUNA contribution (pp) |
|---|---|---|
| A01 | +49.39 | -25.46 |
| A02 | +132.19 | -19.51 |
| A04 | +78.96 | -19.88 |

The legacy LUNA slot contributed **+49.39, +132.19 and +78.96 percentage points** to A01, A02 and A04's development net returns, respectively, but negative contributions in validation. These are NAV-weighted accounting contributions to starting capital, not standalone coin returns. The table reports percentage-point (pp) contributions.

All four selected baskets also lost **before modeled fees** in validation: gross compounded returns ranged from **–15.03% to –29.34%**. Costs deepened the losses rather than explaining them in full.

The diagnostic supports **concentration and period-dependence concerns**. It does not establish that deleting LUNA would have produced the same remaining positions or returns. Documented halts, forced exits and symbol lifecycle treatment were preserved; no event was removed to improve the backtest.

# 5. Relative rank transfer was stronger than economic transfer

The full grid retained **1,117,935 basket–configuration pairs**: 6,615 baskets × 169 settings. Development-ranked groups were frozen before full-grid validation; the 10 accepted case-baskets retain their audited saved paths.

Q1 retained a higher validation median Sharpe than lower-ranked groups, but its median net return remained negative. Much of the rank hierarchy reflects the persistent cost burden of fast settings, not proof that a positive-alpha winner transfers.

These medians are constituent distributions, **not portfolios**. A further 1,000 development all-cash cases had undefined Sharpe and remain unranked in the saved results rather than being assigned to the worst group.

Frozen development-ranked groups

### 2022–2023 development

| Group | Cases | Median net | Median Sharpe | Median annual TO | Positive fraction |
|---|---|---|---|---|---|
| Q1 (best) | 224,867 | -1.55% | 0.14 | 60.01× | 47.19% |
| Q2 | 223,954 | -37.25% | -0.74 | 70.25× | 0.08% |
| Q3 | 219,294 | -69.06% | -1.77 | 159.44× | 0.00% |
| Q4 | 223,910 | -92.90% | -4.03 | 487.59× | 0.00% |
| Q5 (worst) | 224,910 | -99.9973% | -15.69 | 2753.01× | 0.00% |

### 2024–2025 validation

| Group | Cases | Median net | Median Sharpe | Median annual TO | Positive fraction |
|---|---|---|---|---|---|
| Q1 (best) | 224,867 | -16.27% | -0.15 | 64.77× | 30.73% |
| Q2 | 223,954 | -35.59% | -0.76 | 69.73× | 12.99% |
| Q3 | 219,294 | -59.56% | -1.63 | 162.09× | 2.68% |
| Q4 | 223,910 | -90.15% | -4.18 | 489.30× | 0.01% |
| Q5 (worst) | 224,910 | -99.9977% | -17.87 | 2654.28× | 0.00% |

## 6. A high rank need not imply a positive return

Across the full grid, **62.90% of strictly positive development results became strictly negative** in validation. Conversely, **6.40% of strictly negative development results became positive**. The denominators differ, so these are conditional transition rates rather than symmetric probabilities. Exact zeros remain separate.

The best development-ranked group had **224,867 cases**; **69,108 (30.73%)** were profitable in validation. Of these, **39,332** also had positive development returns. The remaining **29,776** were development-negative cases that survived validation. These counts are not independent replications.

### How often the return sign changed

| Family | Dev positive | Positive → negative | Dev negative | Negative → positive |
|---|---|---|---|---|
| All families | 106,302 | 62.90% | 1,010,633 | 6.40% |
| Momentum | 49,083 | 72.29% | 466,887 | 8.69% |
| Reversal | 13,497 | 90.91% | 502,473 | 1.79% |
| Buffered momentum | 10,719 | 66.24% | 15,741 | 35.04% |
| Long/cash trend | 14,898 | 18.53% | 11,562 | 63.81% |
| Long/short trend | 17,355 | 49.43% | 9,105 | 24.91% |
| BTC residual reversal | 750 | 90.40% | 4,865 | 0.00% |

### Development–validation Sharpe rank correlation

| Comparison | Cohorts | Median Spearman | 10th percentile | 90th percentile |
|---|---|---|---|---|
| Across settings within basket | 6,615 | 0.91 | 0.86 | 0.94 |
| Across baskets within setting | 169 | 0.14 | -0.23 | 0.62 |

The median rank correlation was **0.91 across settings within a basket**, but only **0.14 across baskets within the same setting**. That distinction explains why broad setting hierarchies can persist while the exact development-winning basket is unstable.

No IID p-value or confidence interval is attached to these full-grid cross-sectional summaries. Adjacent baskets share seven coins; configurations share signals, prices and dates. Long/cash trend's lower sign-flip rate is descriptive, not a new final-test endorsement.

# 7. Track B was an honest validation-informed selection

After validation had been inspected, a separate rule was frozen **before executing its selection**. Start with development Q1, require evaluated statuses, finite defined Sharpes and strictly positive net returns in both development and validation at 20 bps.

Within each of the existing six families, select the highest validation Sharpe. Ties use lower validation full-L1 turnover, lexical experiment ID and numeric basket ID. Select none when no case qualifies. No positive-Sharpe threshold, annual consistency, stress-cost survival, neighborhood success or risk cap was added as a hidden eligibility gate.

### Every family was retained, including the empty one

| ID | Family | Full grid | Dev Q1 | Eligible | Selected |
|---|---|---|---|---|---|
| B01 | Buffered momentum | 26,460 | 20,068 | 3,619 | 1 |
| B02 | BTC residual reversal | 6,615 | 1,355 | 0 | 0 |
| B03 | Momentum | 515,970 | 107,412 | 13,585 | 1 |
| B04 | Reversal | 515,970 | 45,673 | 1,227 | 1 |
| B05 | Long/cash trend | 26,460 | 24,856 | 12,125 | 1 |
| B06 | Long/short trend | 26,460 | 25,503 | 8,776 | 1 |

### B02 was never a failed selected strategy

B02 is the reserved family label for BTC-factor residual reversal. **Zero** of its **1,355** development-Q1 pairs met the positive-in-both-periods rule, so no B02 candidate was selected. The earlier factor reference control was a diagnostic comparison, not a Track B selection. An empty family is a valid result; it is not replaced by another family or treated as a zero-return investment.

All five selected validation Sharpes are selection statistics from a large correlated pool. They are not unbiased final-test Sharpe estimates, and do not retroactively repair Track A.

# 8. Track B's strong validation results came with weak development cases

B01 and B03 barely cleared the positive-development-return gate. They were eligible under the frozen rule; annual consistency was diagnostic only. Their validation improvement deserves investigation without pretending it was forecast by strong development performance.

### Frozen Track B cases, unchanged in each period

#### 2022–2023 development

| ID | Net return | Sharpe | Annual L1 turnover | Max drawdown |
|---|---|---|---|---|
| B01 | +0.34% | 0.14 | 81.69× | 37.34% |
| B03 | +1.05% | 0.15 | 27.75× | 27.64% |
| B04 | +44.11% | 0.63 | 21.31× | 61.98% |
| B05 | +12.60% | 0.35 | 11.88× | 36.50% |
| B06 | +35.86% | 0.55 | 22.88× | 48.23% |

#### 2024–2025 validation: selection statistics

| ID | Net return | Sharpe | Annual L1 turnover | Max drawdown |
|---|---|---|---|---|
| B01 | +120.19% | 1.62 | 77.45× | 15.50% |
| B03 | +151.30% | 1.85 | 24.13× | 22.08% |
| B04 | +73.78% | 1.14 | 22.34× | 23.42% |
| B05 | +152.00% | 1.40 | 12.48× | 32.48% |
| B06 | +164.24% | 1.24 | 24.59× | 32.83% |

### Annual consistency was not a Track B gate

| ID | 2022 | 2023 | 2024 | 2025 |
|---|---|---|---|---|
| B01 | +8.87% | -7.84% | +48.93% | +47.85% |
| B03 | -9.01% | +11.06% | +26.72% | +98.31% |
| B04 | +36.70% | +5.42% | +46.66% | +18.49% |
| B05 | -33.96% | +70.51% | +111.32% | +19.25% |
| B06 | +8.59% | +25.11% | +56.17% | +69.20% |

# 9. Nine exact identities were frozen for evaluation

BTC and ETH are fixed in every basket; the table lists the other six slots. h is trailing lookback hours, r is rebalance interval hours, and b025 is the whole-target full-L1 buffer of 0.25. Weekly r168 decisions occur Monday at 00:00 UTC; r072 uses the established elapsed-time schedule.

Track A remains the original development-selected procedure; Track B remains the development-plus-validation selection. The shortlist is **not a portfolio**, has no ensemble weights, and does not designate a winner.

### Case–basket identity register

### Track A: four original selections

| ID | Strategy | Basket | Other six coins |
|---|---|---|---|
| A01 | buffered_momentum_h0672_r168_b025 | 4,645 | SOL, LUNA, FTM, TRX, LINK, LTC |
| A02 | momentum_h0672_r072 | 4,645 | SOL, LUNA, FTM, TRX, LINK, LTC |
| A03 | reversal_h1440_r168 | 1,159 | SHIB, DOT, ADA, MANA, FIL, VET |
| A04 | trend_long_short_h0672_r024 | 4,444 | SOL, LUNA, AVAX, FTM, FIL, VET |

### Track B: five validation-informed selections

| ID | Strategy | Basket | Other six coins |
|---|---|---|---|
| B01 | buffered_momentum_h0672_r024_b025 | 2,323 | BNB, XRP, DOGE, FTM, AXS, VET |
| B03 | momentum_h2160_r072 | 2,457 | BNB, XRP, FTM, MATIC, AXS, LTC |
| B04 | reversal_h2160_r168 | 6,064 | DOT, LUNA, AVAX, MANA, FIL, VET |
| B05 | trend_long_cash_h0672_r168 | 2,330 | BNB, XRP, DOGE, TRX, GALA, LINK |
| B06 | trend_long_short_h0672_r168 | 2,495 | BNB, XRP, TRX, MANA, GALA, LINK |

## 10. What was frozen, and what these results can support

**Do not move windows after inspecting performance.** Development and validation each span two complete years, with no result-driven change of boundaries. Annual slices inherit the continuous path's positions and fees; there is no invented internal-year liquidation or cash reset.

**Report the right Sharpe.** Development Sharpe describes fitted evidence; Track A validation Sharpe describes transfer of a development-selected rule; Track B validation Sharpe is selection-biased. The 2026 evaluation must retain this selection history; later explanatory diagnostics are not a new independent test.

**Separate statistical aims.** Track A asks whether a development-selection procedure generalizes. Track B asks whether a strategy selected using development and validation transfers to a later period. Neither requires the development winner to remain rank one, but economic and risk expectations must be stated before that later evaluation.

**Reserved evaluation protocol.** The nine identities and approved risk limits remain fixed. The later approved amendment uses 1 January–30 September 2026 (273 days), rather than a complete calendar year. It requires daily-mean t-statistics, block-bootstrap intervals, multiple-testing correction and honest partial-year labels. No 2026 results are reproduced in this report.

### Common measurement and limitations

**Data source and complete study dataset.** Binance public Data Vision (**data.binance.vision**) spot-USDT **1-hour klines** supply prices and quote volumes. The normalized UTC grid has **43,824 hourly close endpoints**, from **1 October 2021 01:00 to 1 October 2026 00:00**, including warm-up. The packaged **22-slot screening universe** includes SAND; the nine frozen strategies use the **21-slot union of the full universe except SAND**. Lifecycle and missingness masks remain explicit. Data documentation | Full coin inventory | File hashes and coverage.

Net return compounds the saved hourly gross P&L less one-way fee × full-L1 target change, including actual entry and terminal liquidation. Annual turnover is 365 × mean daily full-L1 target change, **not** round-trip turnover or drift-adjusted executed notional. Maximum drawdown is positive peak-to-trough loss magnitude; annualized daily Sharpe uses sample SD, 365 days and zero risk-free rate.

The sensitivity set is 0/5/10/15/40 bps around the 20 bps primary scenario, plus 20 bps with one additional execution hour. Actual borrowing, funding, margin, order-book depth, impact and ordinary drift-maintenance trading were not modeled. Spot shorts are theoretical; historical liquidity and surviving economic slots do not remove universe or lifecycle bias.

Performance tables and fixed-basket attribution describe the same frozen strategies. Source menus expose each component's exact evidence. Detailed configurations, failed cases, cost scenarios, annual risk tables, model logs and provenance are supporting results, not additional reports.

# Per-coin attribution: what each coin contributed, and what joint basket exposure explains

**1 January 2024-31 December 2025, all nine unchanged frozen strategies.** These explanatory diagnostics separate accounting contribution from statistical sensitivity. Strategy returns, Sharpe, turnover, drawdowns, baskets and decisions remain fixed. Track B validation alpha/beta and attribution remain selection-conditioned diagnostics; they are not unbiased final-test results.

The overview shows **annual alpha in percent per year**, estimate [95% CI]. Alpha is 365 times the daily arithmetic intercept, **not CAGR**. The BTC-only comparison uses its full-calendar HAC7 estimate; matched BTC and the joint model use the identical complete-case dates with calendar HAC27.

## 2024-2025 validation: all nine frozen strategies

| ID | Joint alpha / yr [95% CI] | BTC alpha / yr [95% CI] | Matched BTC alpha / yr [95% CI] | Fit / calendar days | $R^2$ | Max VIF |
|---|---|---|---|---|---|---|
| A01 | -8.24 [-37.08, 20.60] | -6.56 [-42.44, 29.33] | -7.07 [-40.32, 26.18] | 727/731 | 0.236 | 3.87 |
| A02 | -16.12 [-51.57, 19.32] | -15.33 [-54.61, 23.95] | -15.70 [-53.27, 21.87] | 727/731 | 0.208 | 3.87 |
| A03 | 4.47 [-17.92, 26.86] | -8.40 [-39.49, 22.69] | -8.40 [-36.24, 19.44] | 731/731 | 0.282 | 5.58 |
| A04 | -1.65 [-85.25, 81.96] | 2.01 [-75.00, 79.02] | 4.59 [-74.54, 83.71] | 727/731 | 0.127 | 3.72 |
| B01 | 19.03 [-14.44, 52.50] | 46.71 [10.96, 82.45] | 45.14 [9.41, 80.87] | 727/731 | 0.176 | 4.25 |
| B03 | 11.95 [-17.16, 41.07] | 52.23 [19.59, 84.87] | 48.75 [17.38, 80.11] | 723/731 | 0.223 | 4.34 |
| B04 | 39.98 [11.80, 68.17] | 31.58 [-4.94, 68.10] | 31.58 [-4.17, 67.34] | 731/731 | 0.308 | 5.37 |
| B05 | 31.81 [1.36, 62.25] | 31.35 [-18.03, 80.72] | 31.35 [-20.51, 83.20] | 731/731 | 0.634 | 3.96 |
| B06 | 70.23 [8.68, 131.77] | 70.06 [0.22, 139.89] | 70.06 [3.93, 136.19] | 731/731 | 0.101 | 3.83 |

**How to read this.** The joint regression includes an intercept and all eight basket coins simultaneously. A conditional beta is sensitivity holding the other seven factors fixed; it is not a portfolio weight, a causal contribution or proof of market neutrality. R-squared is an in-sample descriptive fit. Comparing joint with matched BTC separates added factors from sample exclusions; the full-calendar BTC-only comparison also uses a different covariance convention.

**Uncertainty and missingness.** Joint and matched BTC intervals are approximate marginal 95% normal-Wald intervals using Bartlett calendar HAC27 and n/(n-k) correction; HAC7 sensitivity is exported. Missing factor days are excluded from regressions only; gaps are not filled or compressed. The complete strategy and attribution calendars remain intact. Standardized condition >30 or any VIF >10 is a warning, not a reason to drop a coin. Some fits are sensitive to extreme days; high leverage/Cook's distance can coexist with modest collinearity. No event is removed and no model reselected.

**Per-coin accounting.** Gross contributions and positive allocated fees are both linked with prior **net** strategy NAV; net = gross - fees. Units are whole-period percentage points (pp) of initial capital, not standalone coin returns. The next pages retain all eight slots and their marginal net-contribution intervals.

## A01 | basket 4645 | 2024-2025 validation

Buffered momentum; 28.0d lookback, 168h rebalance. **Net period return: -29.55%.** The eight net contributions below sum to that unchanged return.

Joint fit **727/731 days**; R-squared **0.236**; standardized condition **5.04**; max VIF **3.87**. Neither declared collinearity warning threshold was crossed. Max leverage **0.746**, max Cook's distance **12.496**: low collinearity alone does not establish stability. Excluded: **2025-01-13 to 2025-01-16 (4 days)**.

BTC-only beta: full-calendar HAC7 **-0.123 [-0.185, -0.060]**; matched-calendar HAC27 **-0.123 [-0.196, -0.050]**. All brackets are marginal 95% intervals.

### A01: per-coin accounting and conditional betas

| Coin | Gross pp | Fees pp | Net pp | Net 95% CI (pp) | Joint beta [95% CI] |
|---|---|---|---|---|---|
| BTC | -0.17 | 1.23 | -1.40 | [-12.59, 11.10] | 0.014 [-0.087, 0.115] |
| ETH | -1.80 | 1.22 | -3.02 | [-21.45, 23.47] | -0.017 [-0.119, 0.085] |
| SOL | 0.50 | 1.30 | -0.80 | [-26.60, 31.17] | -0.005 [-0.096, 0.086] |
| LUNA/LUNC | -23.94 | 1.52 | -25.46 | [-66.13, 11.32] | -0.112 [-0.163, -0.061] |
| FTM/S | 16.72 | 1.23 | 15.49 | [-26.38, 78.63] | -0.021 [-0.088, 0.047] |
| TRX | -5.05 | 1.55 | -6.60 | [-39.58, 12.80] | -0.092 [-0.157, -0.026] |
| LINK | 7.92 | 1.30 | 6.63 | [-17.41, 31.79] | 0.038 [-0.089, 0.165] |
| LTC | -13.02 | 1.36 | -14.38 | [-32.66, -0.54] | 0.019 [-0.043, 0.081] |

## A02 | basket 4645 | 2024-2025 validation

Momentum; 28.0d lookback, 72h rebalance. **Net period return: -39.63%.** The eight net contributions below sum to that unchanged return.

Joint fit **727/731 days**; R-squared **0.208**; standardized condition **5.04**; max VIF **3.87**. Neither declared collinearity warning threshold was crossed. Max leverage **0.746**, max Cook's distance **13.712**: low collinearity alone does not establish stability. Excluded: **2025-01-13 to 2025-01-16 (4 days)**.

BTC-only beta: full-calendar HAC7 **-0.100 [-0.161, -0.039]**; matched-calendar HAC27 **-0.100 [-0.176, -0.024]**. All brackets are marginal 95% intervals.

### A02: per-coin accounting and conditional betas

| Coin | Gross pp | Fees pp | Net pp | Net 95% CI (pp) | Joint beta [95% CI] |
|---|---|---|---|---|---|
| BTC | 4.58 | 2.00 | 2.58 | [-8.72, 15.18] | 0.064 [-0.044, 0.172] |
| ETH | -3.50 | 2.00 | -5.50 | [-22.54, 19.88] | -0.039 [-0.143, 0.065] |
| SOL | -2.07 | 2.24 | -4.31 | [-27.95, 24.00] | -0.027 [-0.111, 0.056] |
| LUNA/LUNC | -17.19 | 2.32 | -19.51 | [-65.67, 21.11] | -0.101 [-0.147, -0.054] |
| FTM/S | 10.80 | 2.10 | 8.69 | [-29.83, 61.43] | -0.012 [-0.085, 0.060] |
| TRX | -4.62 | 2.62 | -7.23 | [-37.82, 13.52] | -0.094 [-0.164, -0.024] |
| LINK | 9.81 | 2.24 | 7.58 | [-14.91, 34.85] | 0.033 [-0.097, 0.163] |
| LTC | -19.65 | 2.28 | -21.93 | [-41.22, -4.61] | 0.032 [-0.035, 0.099] |

## A03 | basket 1159 | 2024-2025 validation

Reversal; 60.0d lookback, 168h rebalance. **Net period return: -22.31%.** The eight net contributions below sum to that unchanged return.

Joint fit **731/731 days**; R-squared **0.282**; standardized condition **6.44**; max VIF **5.58**. Neither declared collinearity warning threshold was crossed. Max leverage **0.499**, max Cook's distance **2.425**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **-0.026 [-0.083, 0.031]**; matched-calendar HAC27 **-0.026 [-0.089, 0.037]**. All brackets are marginal 95% intervals.

### A03: per-coin accounting and conditional betas

| Coin | Gross pp | Fees pp | Net pp | Net 95% CI (pp) | Joint beta [95% CI] |
|---|---|---|---|---|---|
| BTC | -2.26 | 1.04 | -3.30 | [-18.88, 12.96] | -0.102 [-0.174, -0.029] |
| ETH | 3.51 | 0.85 | 2.66 | [-23.48, 31.37] | 0.010 [-0.085, 0.106] |
| SHIB | -10.60 | 0.98 | -11.58 | [-43.28, 19.02] | -0.025 [-0.078, 0.028] |
| DOT | 13.58 | 1.06 | 12.52 | [-9.85, 40.20] | 0.075 [0.009, 0.140] |
| ADA | -22.43 | 1.04 | -23.46 | [-64.17, 12.61] | -0.114 [-0.151, -0.076] |
| MANA | -12.17 | 1.10 | -13.27 | [-38.20, 8.58] | 0.051 [-0.004, 0.105] |
| FIL | 11.18 | 1.18 | 10.00 | [-28.22, 71.36] | 0.100 [0.038, 0.163] |
| VET | 5.25 | 1.13 | 4.12 | [-20.67, 31.52] | -0.015 [-0.075, 0.046] |

## A04 | basket 4444 | 2024-2025 validation

Trend long/short; 28.0d lookback, 24h rebalance. **Net period return: -45.31%.** The eight net contributions below sum to that unchanged return.

Joint fit **727/731 days**; R-squared **0.127**; standardized condition **5.28**; max VIF **3.72**. Neither declared collinearity warning threshold was crossed. Max leverage **0.489**, max Cook's distance **1.315**: low collinearity alone does not establish stability. Excluded: **2025-01-13 to 2025-01-16 (4 days)**.

BTC-only beta: full-calendar HAC7 **-0.309 [-0.548, -0.071]**; matched-calendar HAC27 **-0.304 [-0.639, 0.031]**. All brackets are marginal 95% intervals.

### A04: per-coin accounting and conditional betas

| Coin | Gross pp | Fees pp | Net pp | Net 95% CI (pp) | Joint beta [95% CI] |
|---|---|---|---|---|---|
| BTC | -1.27 | 3.12 | -4.40 | [-14.79, 20.05] | -0.001 [-0.308, 0.307] |
| ETH | 3.21 | 2.24 | 0.97 | [-11.88, 33.41] | -0.035 [-0.265, 0.196] |
| SOL | 5.49 | 1.89 | 3.60 | [-13.32, 42.91] | -0.183 [-0.311, -0.055] |
| LUNA/LUNC | -17.11 | 2.77 | -19.88 | [-55.52, 11.57] | -0.035 [-0.093, 0.022] |
| AVAX | 3.15 | 2.24 | 0.92 | [-12.20, 54.04] | 0.036 [-0.062, 0.133] |
| FTM/S | -8.48 | 2.61 | -11.08 | [-45.62, 53.22] | -0.058 [-0.133, 0.017] |
| FIL | -10.93 | 2.96 | -13.89 | [-64.05, 22.93] | -0.080 [-0.183, 0.023] |
| VET | 0.20 | 1.74 | -1.55 | [-18.64, 57.38] | 0.118 [-0.035, 0.270] |

## B01 | basket 2323 | 2024-2025 validation

Buffered momentum; 28.0d lookback, 24h rebalance. **Net period return: 120.19%.** The eight net contributions below sum to that unchanged return.

Joint fit **727/731 days**; R-squared **0.176**; standardized condition **6.02**; max VIF **4.25**. Neither declared collinearity warning threshold was crossed. Max leverage **0.272**, max Cook's distance **0.427**: low collinearity alone does not establish stability. Excluded: **2025-01-13 to 2025-01-16 (4 days)**.

BTC-only beta: full-calendar HAC7 **-0.080 [-0.142, -0.017]**; matched-calendar HAC27 **-0.081 [-0.156, -0.005]**. All brackets are marginal 95% intervals.

### B01: per-coin accounting and conditional betas

| Coin | Gross pp | Fees pp | Net pp | Net 95% CI (pp) | Joint beta [95% CI] |
|---|---|---|---|---|---|
| BTC | 8.25 | 5.64 | 2.61 | [-19.92, 44.10] | -0.003 [-0.120, 0.115] |
| ETH | 20.94 | 5.15 | 15.79 | [-16.05, 58.29] | -0.074 [-0.191, 0.043] |
| BNB | -6.58 | 5.34 | -11.92 | [-53.40, 14.90] | 0.065 [0.006, 0.124] |
| XRP | 33.26 | 4.54 | 28.73 | [-26.13, 144.54] | 0.077 [-0.014, 0.167] |
| DOGE | 36.82 | 5.21 | 31.61 | [-24.32, 185.67] | 0.042 [-0.024, 0.109] |
| FTM/S | 33.03 | 5.68 | 27.34 | [-61.95, 119.82] | -0.010 [-0.079, 0.060] |
| AXS | 11.45 | 5.65 | 5.81 | [-57.38, 57.29] | -0.170 [-0.236, -0.104] |
| VET | 25.13 | 4.91 | 20.22 | [-25.96, 81.90] | 0.036 [-0.032, 0.104] |

## B03 | basket 2457 | 2024-2025 validation

Momentum; 90.0d lookback, 72h rebalance. **Net period return: 151.30%.** The eight net contributions below sum to that unchanged return.

Joint fit **723/731 days**; R-squared **0.223**; standardized condition **5.80**; max VIF **4.34**. Neither declared collinearity warning threshold was crossed. Max leverage **0.145**, max Cook's distance **0.422**: low collinearity alone does not establish stability. Excluded: **2024-09-10 to 2024-09-13 (4 days); 2025-01-13 to 2025-01-16 (4 days)**.

BTC-only beta: full-calendar HAC7 **-0.054 [-0.117, 0.008]**; matched-calendar HAC27 **-0.057 [-0.132, 0.017]**. All brackets are marginal 95% intervals.

### B03: per-coin accounting and conditional betas

| Coin | Gross pp | Fees pp | Net pp | Net 95% CI (pp) | Joint beta [95% CI] |
|---|---|---|---|---|---|
| BTC | 11.11 | 1.87 | 9.23 | [-11.50, 42.98] | 0.080 [-0.008, 0.168] |
| ETH | -3.94 | 1.48 | -5.42 | [-54.32, 49.26] | -0.024 [-0.115, 0.068] |
| BNB | 12.99 | 1.94 | 11.06 | [-22.50, 70.77] | 0.088 [-0.033, 0.210] |
| XRP | 15.65 | 1.58 | 14.08 | [-36.66, 111.18] | 0.109 [0.018, 0.199] |
| FTM/S | 81.54 | 1.49 | 80.05 | [-3.97, 199.55] | 0.006 [-0.058, 0.069] |
| MATIC/POL | 28.86 | 1.83 | 27.03 | [-10.41, 93.97] | -0.084 [-0.140, -0.028] |
| AXS | 35.73 | 1.54 | 34.19 | [-26.51, 99.26] | -0.141 [-0.210, -0.072] |
| LTC | -16.77 | 2.14 | -18.91 | [-58.41, 4.28] | -0.016 [-0.121, 0.089] |

## B04 | basket 6064 | 2024-2025 validation

Reversal; 90.0d lookback, 168h rebalance. **Net period return: 73.78%.** The eight net contributions below sum to that unchanged return.

Joint fit **731/731 days**; R-squared **0.308**; standardized condition **6.31**; max VIF **5.37**. Neither declared collinearity warning threshold was crossed. Max leverage **0.497**, max Cook's distance **9.072**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **-0.007 [-0.061, 0.046]**; matched-calendar HAC27 **-0.007 [-0.069, 0.054]**. All brackets are marginal 95% intervals.

### B04: per-coin accounting and conditional betas

| Coin | Gross pp | Fees pp | Net pp | Net 95% CI (pp) | Joint beta [95% CI] |
|---|---|---|---|---|---|
| BTC | -7.23 | 1.09 | -8.32 | [-34.48, 22.61] | -0.119 [-0.187, -0.051] |
| ETH | 31.95 | 1.18 | 30.77 | [-14.00, 93.27] | -0.009 [-0.114, 0.097] |
| DOT | 16.52 | 1.34 | 15.18 | [-15.45, 54.22] | -0.006 [-0.108, 0.097] |
| LUNA/LUNC | -18.15 | 1.89 | -20.04 | [-81.53, 52.33] | 0.095 [0.002, 0.188] |
| AVAX | 18.98 | 1.93 | 17.05 | [-32.13, 68.31] | -0.041 [-0.135, 0.054] |
| MANA | -12.20 | 1.60 | -13.80 | [-53.79, 40.10] | 0.003 [-0.080, 0.087] |
| FIL | 52.05 | 1.72 | 50.33 | [-13.89, 178.01] | 0.144 [0.052, 0.236] |
| VET | 4.21 | 1.60 | 2.61 | [-42.44, 49.42] | -0.074 [-0.125, -0.024] |

## B05 | basket 2330 | 2024-2025 validation

Trend long/cash; 28.0d lookback, 168h rebalance. **Net period return: 152.00%.** The eight net contributions below sum to that unchanged return.

Joint fit **731/731 days**; R-squared **0.634**; standardized condition **5.20**; max VIF **3.96**. Neither declared collinearity warning threshold was crossed. Max leverage **0.754**, max Cook's distance **0.635**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **0.462 [0.361, 0.562]**; matched-calendar HAC27 **0.462 [0.318, 0.606]**. All brackets are marginal 95% intervals.

### B05: per-coin accounting and conditional betas

| Coin | Gross pp | Fees pp | Net pp | Net 95% CI (pp) | Joint beta [95% CI] |
|---|---|---|---|---|---|
| BTC | 10.99 | 1.31 | 9.69 | [-4.60, 81.36] | -0.070 [-0.217, 0.078] |
| ETH | 21.47 | 1.15 | 20.31 | [-6.46, 125.01] | -0.002 [-0.104, 0.099] |
| BNB | 26.05 | 1.35 | 24.69 | [-2.87, 114.62] | 0.070 [0.003, 0.137] |
| XRP | 29.09 | 1.03 | 28.06 | [-12.53, 220.06] | 0.059 [-0.011, 0.129] |
| DOGE | 12.72 | 1.11 | 11.61 | [-21.25, 200.25] | 0.121 [0.077, 0.166] |
| TRX | 19.46 | 1.07 | 18.38 | [-3.18, 118.96] | 0.107 [0.089, 0.124] |
| GALA | 22.16 | 0.82 | 21.34 | [-11.29, 196.47] | 0.086 [0.055, 0.117] |
| LINK | 18.77 | 0.85 | 17.91 | [-13.84, 109.17] | 0.047 [-0.019, 0.114] |

## B06 | basket 2495 | 2024-2025 validation

Trend long/short; 28.0d lookback, 168h rebalance. **Net period return: 164.24%.** The eight net contributions below sum to that unchanged return.

Joint fit **731/731 days**; R-squared **0.101**; standardized condition **5.41**; max VIF **3.83**. Neither declared collinearity warning threshold was crossed. Max leverage **0.753**, max Cook's distance **0.313**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **-0.203 [-0.394, -0.012]**; matched-calendar HAC27 **-0.203 [-0.482, 0.076]**. All brackets are marginal 95% intervals.

### B06: per-coin accounting and conditional betas

| Coin | Gross pp | Fees pp | Net pp | Net 95% CI (pp) | Joint beta [95% CI] |
|---|---|---|---|---|---|
| BTC | 7.63 | 2.28 | 5.34 | [-14.89, 61.28] | -0.229 [-0.509, 0.051] |
| ETH | 23.99 | 2.03 | 21.96 | [-11.60, 128.28] | -0.115 [-0.334, 0.104] |
| BNB | 18.85 | 2.42 | 16.43 | [-12.50, 97.53] | 0.016 [-0.124, 0.155] |
| XRP | 14.54 | 1.83 | 12.71 | [-30.79, 158.68] | 0.009 [-0.110, 0.127] |
| TRX | 11.63 | 1.79 | 9.84 | [-11.24, 92.69] | 0.106 [0.064, 0.149] |
| MANA | 30.65 | 1.54 | 29.11 | [-13.82, 168.05] | 0.070 [-0.057, 0.197] |
| GALA | 53.80 | 1.36 | 52.43 | [0.34, 259.05] | 0.046 [-0.053, 0.145] |
| LINK | 17.95 | 1.53 | 16.41 | [-29.09, 104.59] | -0.043 [-0.173, 0.086] |

## Full precision, sensitivity and multiplicity

Per-coin gross, fee and net attribution with all three confidence intervals | Joint alpha/betas, HAC7 sensitivity, global Holm243 p-values and Bonferroni243 intervals | Joint model diagnostics and individual VIFs | Full-calendar BTC-only estimates and intervals | Matched BTC-only coefficients and intervals | Regression calendar and exclusions.

Attribution uses **4,000 seeded, synchronized circular 28-day calendar-block draws** across all nine strategies and eight coins. Daily gross/fee vectors are resampled together and relinked using each draw's **new prior NAV**; signals, targets and trades are not regenerated and no block-boundary fee is invented. Any invalid draw makes the affected interval unavailable; none is silently dropped. Gross/fee intervals and invalid-draw counts remain in the CSV.

The coefficient correction spans **243 predeclared diagnostic slots: 27 joint alphas + 216 joint betas** across all three periods. Displayed intervals are marginal; approximate Bonferroni familywise intervals and Holm-adjusted p-values are exported. This corrects only that conditional coefficient family, **not the original universe, configuration, period or strategy-selection search**. Attribution intervals are marginal and not multiplicity-adjusted. No significance row changes the frozen shortlist or crowns a new winner. LUNA/LUNC is legacy Terra, not Terra 2.0; FTM/S and MATIC/POL retain the original economic slots.

# Performance above a 3% nominal cash benchmark

**2024-2025 validation; all nine unchanged strategies.** Cash accrues at c = 0.03/365 each UTC calendar day (ACT/365F), compounded daily: **6.19% over 731 days**; the 365-day effective return is 3.0453%. This is a hypothetical constant **USDT cash proxy**, not an actual yield product; a USD interpretation assumes USDT/USD parity. **No interest is credited to strategy cash or short collateral.**

## Raw returns versus cash-relative performance; drawdown uses DAILY endpoints

| ID | Raw net return | Return vs cash (wealth-relative) | Relative CAGR | Excess Sharpe [95% CI] | Relative daily Max DD |
|---|---|---|---|---|---|
| A01 | -29.55% | -33.66% | -18.53% | -0.49 [-1.49, 0.59] | 46.22% |
| A02 | -39.63% | -43.15% | -24.57% | -0.73 [-1.81, 0.43] | 49.65% |
| A03 | -22.31% | -26.84% | -14.45% | -0.52 [-1.85, 0.51] | 44.91% |
| A04 | -45.31% | -48.50% | -28.20% | -0.27 [-1.63, 1.05] | 69.47% |
| B01 | 120.19% | 107.35% | 43.92% | 1.51 [0.29, 2.76] | 14.71% |
| B03 | 151.30% | 136.65% | 53.74% | 1.73 [0.57, 2.87] | 21.18% |
| B04 | 73.78% | 63.65% | 27.88% | 1.03 [-0.17, 2.03] | 21.78% |
| B05 | 152.00% | 137.31% | 53.96% | 1.33 [-0.42, 3.03] | 31.55% |
| B06 | 164.24% | 148.84% | 57.65% | 1.18 [-0.07, 2.45] | 29.33% |

## Cash-excess alpha: percent per year, not CAGR; intervals are marginal

| ID | BTC-only alpha/yr [95% CI], full HAC7 | Matched BTC alpha/yr [95% CI], HAC27 | Joint basket alpha/yr [95% CI], HAC27 | Mean-excess HAC27 t |
|---|---|---|---|---|
| A01 | -9.92% [-45.75%, 25.90%] | -10.44% [-43.61%, 22.73%] | -11.77% [-40.54%, 17.00%] | -0.90 |
| A02 | -18.63% [-57.87%, 20.61%] | -19.00% [-56.52%, 18.52%] | -19.56% [-54.96%, 15.84%] | -1.23 |
| A03 | -11.47% [-42.57%, 19.62%] | -11.47% [-39.29%, 16.34%] | 1.41% [-20.97%, 23.79%] | -0.90 |
| A04 | -1.91% [-78.94%, 75.11%] | 0.68% [-78.74%, 80.09%] | -5.36% [-89.32%, 78.60%] | -0.40 |
| B01 | 43.47% [7.72%, 79.22%] | 41.90% [6.10%, 77.70%] | 15.92% [-17.52%, 49.37%] | 2.24 |
| B03 | 49.07% [16.47%, 81.67%] | 45.57% [14.24%, 76.90%] | 9.01% [-20.06%, 38.07%] | 2.90 |
| B04 | 28.56% [-7.91%, 65.02%] | 28.56% [-7.14%, 64.26%] | 36.96% [8.79%, 65.13%] | 1.59 |
| B05 | 29.73% [-19.65%, 79.12%] | 29.73% [-22.24%, 81.71%] | 30.06% [-0.42%, 60.54%] | 1.39 |
| B06 | 66.45% [-3.37%, 136.26%] | 66.45% [0.15%, 132.74%] | 66.81% [5.17%, 128.44%] | 1.71 |

**Definitions.** Return vs cash = W(strategy)/W(cash) − 1, not the difference of cumulative returns or a compounded arithmetic excess. Relative CAGR annualizes this wealth ratio. Excess Sharpe uses daily r − c, sample standard deviation and $\sqrt{365}$. Relative daily drawdown includes initial NAV and is **not comparable to the original hourly Max DD** as evidence of improved risk.

**Inference.** Sharpe intervals use 4,000 synchronized circular 28-day calendar-block draws, seed 20261016; return/growth intervals are exported. Mean-excess t/95% intervals use calendar HAC27 and n/(n−1); a **new Holm27 family** spans nine strategies × three periods. Factor regressions subtract c from both strategy and all coin returns, retaining their original sample: joint/matched BTC use HAC27 with n/(n−k); full-calendar BTC uses HAC7 without correction. Alpha intervals are recalculated; betas and beta intervals are unchanged. A **separate new cash-coefficient Holm243/Bonferroni243 family** is exported; raw-return corrections remain intact.

**Scope.** These conditional diagnostics are not adjusted for the original basket/configuration/selection search. Frozen returns, trades, turnover, selections, raw-return risk limits and acceptance flags remain unchanged. No financing model or new winner is introduced.

**Full precision, all three periods:** Performance and bootstrap intervals | Combined summary | Mean tests and Holm27 | Joint coefficients and cash Holm243/Bonferroni243 | Matched BTC | Full-calendar BTC | Model samples.